

1 / 5 | Ages 7-8 | 15 minutes

The twenty-token shop

What we are practising

Add prices and subtract spending from a fixed total.

What you need

20 large paper tokens / Three household objects / Paper price labels

Draw tokens and cross them out instead of moving pieces.

Before you begin

Label a book 7 tokens, a cup 4 and a cloth 3; allow repeated purchases.

Use large paper tokens, not real coins, when younger children are nearby.

Try it together

1. Count a budget of 20 tokens.
2. Buy one of each item and work out the total.
3. Find the change, then choose something affordable with it.
4. Find a different basket costing exactly 20.

Talk, draw or show your idea

Can two different baskets have the same total?

Space to explore

2 / 5 | Ages 7-8 | 15 minutes

Fraction picnic

What we are practising

Recognize equivalent shares using halves and quarters.

What you need

Two equal rectangular sheets / Pencil

Draw two same-size rectangles and mark equal sections.

Before you begin

Call each sheet one whole picnic sandwich.

Use paper models; no food or cutting is required.

Try it together

1. Fold the first sheet into two equal parts and shade one.
2. Fold the second into four equal parts and shade two.
3. Overlay or line up the shaded amounts and compare.
4. Explain how two people could share one whole fairly.

Talk, draw or show your idea

Why must the starting wholes be the same size?

Space to explore

3 / 5 | Ages 7-8 | 15 minutes

Measure twice

What we are practising

Measure a length with repeated equal units and explain unit size.

What you need

A short pencil / A longer pencil / Paper

Use two different lengths of folded paper.

Before you begin

Choose a reachable table edge and label the two measuring units A and B.

Measure reachable surfaces without climbing; keep pencil points away from faces.

Try it together

1. Predict which pencil will need more end-to-end placements.
2. Move the short pencil along the edge without gaps, counting whole placements and noting any remainder.
3. Repeat with the longer pencil along the same edge.
4. Record each result with its unit name.

Talk, draw or show your idea

Did the table change, or did the unit change?

Space to explore

4 / 5 | Ages 7-8 | 10 minutes

Timetable detective

What we are practising

Add minutes across an hour boundary.

What you need

Paper / Pencil

Use a clock face drawn on paper.

Before you begin

Write a start time of 3:40, followed by 15 minutes reading and 10 minutes drawing.

Keep this untimed; use invented times if schedules feel stressful.

Try it together

1. Mark the starting time on a simple timeline.
2. Add the reading time and record its end.
3. Add the drawing time and record the final time.
4. Check by counting the total elapsed minutes.

Talk, draw or show your idea

How many drawing minutes happen before 4:00?

Space to explore

5 / 5 | Ages 7-8 | 20 minutes

A tiny survey

What we are practising

Use tally marks and compare counts on a common scale.

What you need

Paper / Pencil

Use an invented set of six votes instead of asking people.

Before you begin

Choose three activity options: drawing, stories and building.

Participation is optional; record votes without names or personal details.

Try it together

1. Ask willing family members to choose one option each, or use votes: drawing, stories, drawing, building, stories, drawing.
2. Record one tally for each vote.
3. Draw three bars with one equal-sized square per vote.
4. Label the choices and counts, then compare the tallest and shortest bars.

Talk, draw or show your idea

Does this small survey tell us what every child prefers?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. The twenty-token shop

A little hint

One of each costs $7 + 4 + 3$; two books leave 6 tokens.

Answer / what to notice

One of each costs 14, leaving 6. Two books and two cloths cost 20.

Add all prices before subtracting; a purchase is affordable if its price does not exceed the balance.

Make it easier

Use a budget of 10 and prices 2 and 5.

Take it further

Find every basket of these goods that totals 20.

Another day, away from the screen

Make a pretend shop with new whole-number prices.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Fraction picnic

A little hint

Two quarters cover the same area as one half of the same whole.

Answer / what to notice

One half equals two quarters; each of two people gets one half.

The number of parts changes, but the total shaded amount stays the same when the wholes match.

Make it easier

Compare only one whole with two halves.

Take it further

Fold another equal sheet into eighths and find the equivalent half.

Another day, away from the screen

Look for halves and quarters in folded napkins.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Measure twice

A little hint

A smaller unit usually fits more times into a fixed length.

Answer / what to notice

Measurements depend on the objects used; the shorter unit needs at least as many placements.

Equal units placed without gaps make a repeatable measurement; unit names make the number meaningful.

Make it easier

Measure a short book edge using one unit.

Take it further

Compare both measurements with a ruler and discuss leftover fractions.

Another day, away from the screen

Measure three safe objects using one shared unit and order their lengths.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Timetable detective

A little hint

Reading ends at 3:55; five more minutes reach 4:00.

Answer / what to notice

Reading ends at 3:55 and drawing at 4:05; five drawing minutes come before 4:00.

The total is 25 minutes, and 20 minutes take the clock from 3:40 to 4:00.

Make it easier

Start at 3:00 so both activities finish within the same hour.

Take it further

Add a 7-minute break and find the new ending time.

Another day, away from the screen

Plan two real activities and compare estimated with actual durations.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. A tiny survey

A little hint

Check that all bar counts add up to the number of votes.

Answer / what to notice

The sample votes give drawing 3, stories 2 and building 1; drawing leads building by 2.

A chart summarizes the people or sample surveyed, not everyone everywhere.

Make it easier

Compare only two choices.

Take it further

Show the same data in a table and explain which display is easier to read.

Another day, away from the screen

Survey a different harmless preference and compare the totals.